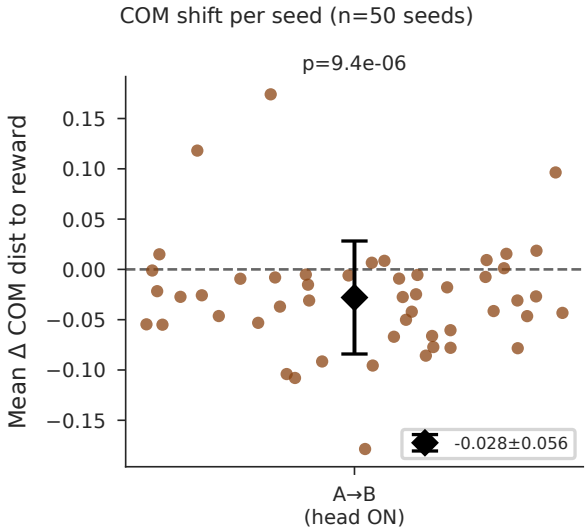
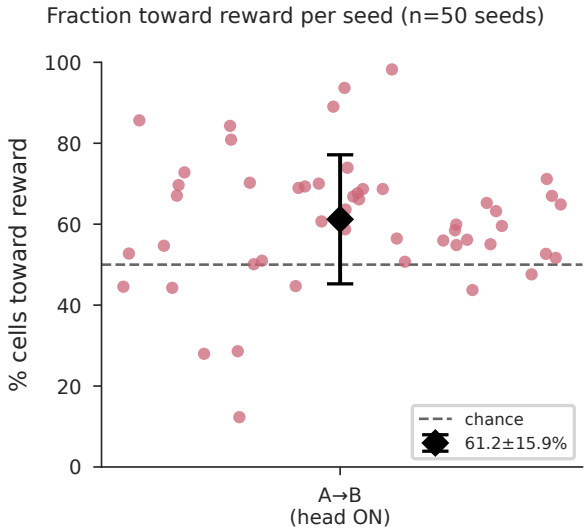
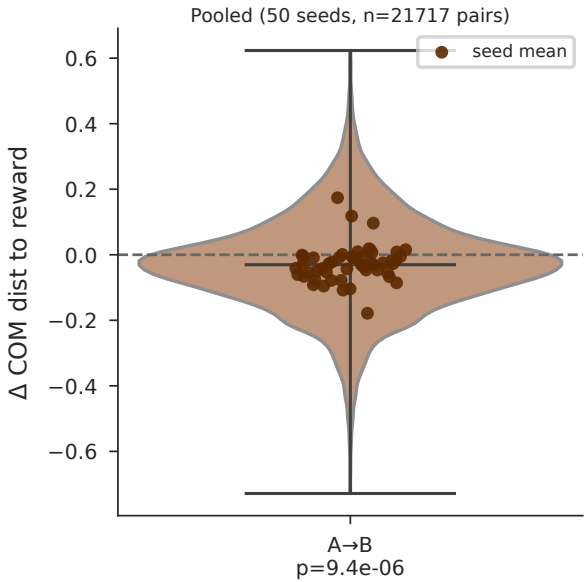
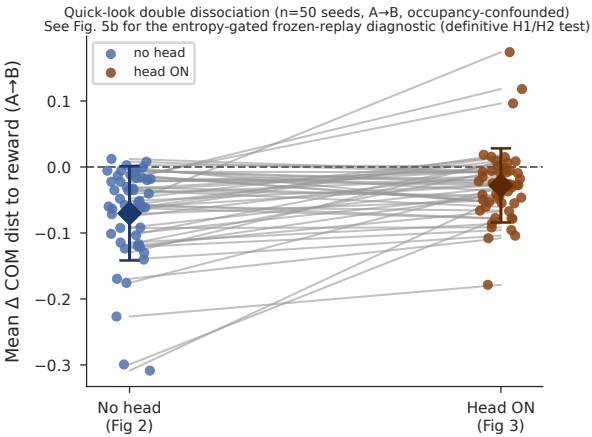

COM shift per seed (n=50 seeds)
p=9.4e-06
Mean Δ COM dist to reward
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
-0.028±0.056
A→B
(head ON)


Fraction toward reward per seed (n=50 seeds)
% cells toward reward
100
80
60
40
20
0
chance
61.2±15.9%
A→B
(head ON)


Pooled (50 seeds, n=21717 pairs)
seed mean
Δ COM dist to reward
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
A→B
p=9.4e-06


Quick-look double dissociation (n=50 seeds, A→B, occupancy-confounded)
See Fig. 5b for the entropy-gated frozen-replay diagnostic (definitive H1/H2 test)
no head
head ON
Mean Δ COM dist to reward (A→B)
0.1
0.0
−0.1
−0.2
−0.3
No head
(Fig 2)
Head ON
(Fig 3)